

Provenance-Gated AMM-Flow Co-Movement and Predictive Benchmarking During Stablecoin Stress

Evidence from Curve Finance Logs and a Multi-Event ML Prediction Framework

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ABSTRACT

Stablecoin stress episodes are price dislocations, but they are also *liquidity-flow events*: traders swap through AMM pools, redeem stablecoins through mint-and-burn channels, and rebalance across venues in ways fully observable from on-chain logs. We present a provenance-aware benchmark framework that assigns Tier-A (execution-grade on-chain) or Tier-B (public market context) status to every node and feature, then filters each empirical edge through a provenance gate and a statistical gate before allowing a paper-level claim. We make two contributions. First, across five historical stress episodes, the only robust paper-claimable A/A result is in the USDT/Curve 2023 event: the Curve 3pool and Curve crvUSD/USDT pool exhibit statistically supported bidirectional AMM-flow co-movement ($\hat{\rho} = 0.386$, 95% CI [0.25, 0.51], $n = 168$ hourly observations, Bonferroni $p \leq 0.014$) using Tier-A `usdc_net_sold_1h` data. Second, a four-classifier ML benchmark trained under leave-one-event-out evaluation achieves AUROC = 0.859 (Random Forest) on USDC/SVB 2023 for downstream price-stress prediction; provenance-gated Tier-A AMM-flow features contribute a +5.1 pp AUROC lift over a Tier-B-only baseline on the best ensemble model. We release the benchmark with provenance manifests and reproducible event windows.

CCS CONCEPTS

• Applied computing → Economics; • Computing methodologies → Machine learning.

KEYWORDS

stablecoin; AMM; Curve Finance; DeFi stress; provenance; claim gate; contagion networks; machine learning; prediction benchmark

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1 INTRODUCTION

Between May 2022 and March 2023, four distinct stablecoin systems experienced significant stress: the algorithmic UST/Terra collapsed; FTX's implosion triggered exchange runs; USDC briefly de-pegged following Silicon Valley Bank's failure; and Curve Finance pools became severely imbalanced in June 2023. Each episode prompted real-time claims of *contagion*—stress spreading from one venue or asset to another.

Most empirical analyses of these events rely on price data alone and treat all data sources as equally reliable. This paper addresses both limitations by introducing a provenance-aware framework that makes data quality an explicit input to every empirical claim, and by pairing the co-movement analysis with a machine learning prediction benchmark that quantifies the informational value of Tier-A on-chain features relative to Tier-B market proxies.

Contribution 1: Stress as a flow event. When a stablecoin de-pegs, market participants do not merely observe a price change—they execute swaps in automated market maker (AMM) pools, burn or mint stablecoins through issuer channels, and rebalance across venues. These flows are *completely and immutably observable* from on-chain event logs. Curve Finance's TokenExchange logs record every swap with exact block timestamps, exact amounts, and on-chain provenance. This makes Curve AMM-flow data execution-grade in a way that public centralized exchange (CEX) price feeds are not.

Contribution 2: A provenance-aware claim gate. We introduce a three-gate pipeline that filters every empirical edge by (i) data quality tier, (ii) feature-level tier, and (iii) statistical significance before granting a paper-level claim. Historical full-depth CEX order books—tick-by-tick L2 data from Binance, Coinbase, or Kraken—are not freely available for any of the five episodes studied here. Consequently, any claim about historical CEX execution-grade microstructure transmission is explicitly blocked by our gate.

Contribution 3: A provenance-gated ML prediction benchmark. We train four classifiers (Random Forest, LightGBM, XGBoost, Logistic Regression) to predict downstream stablecoin price stress using features partitioned by provenance tier. A leave-one-event-out evaluation on the USDC/SVB 2023 episode yields AUROC = 0.859 (Random Forest, full features) versus AUROC = 0.808 with Tier-B features only—a +5.1 pp lift attributable to Tier-A AMM-flow features. The benchmark formalises the provenance gate as a reusable ML infrastructure: Tier-A features are not just methodologically cleaner, they are informationally richer.

Main co-movement finding. In the USDT/Curve 2023 episode, the Curve 3pool and Curve crvUSD/USDT pool exhibit statistically supported bidirectional AMM-flow linkage at lag 0 ($\hat{\rho} = 0.386$, 95% CI [0.25, 0.51], Bonferroni $p \leq 0.014$) using Tier-A hourly `usdc_net_sold_1h` data. Terra/LUNA has provenance-valid A/A

candidates but fails the statistical gate at the hourly grid. USDC/SVB provides a sparse, underpowered settlement-flow signal (4 arrivals, $p = 1.00$). FTX and BUSD yield A/B contextual evidence only.

The paper proceeds as follows. Section 2 reviews related work. Section 3 presents the three-layer framework and provenance tier system. Section 4 describes the claim-gated methodology. Sections 5–8 report the empirical co-movement results. Section 7 presents the ML prediction benchmark. Section 9 addresses robustness and scope. Section 10 concludes.

2 RELATED WORK

Stablecoin stability and DeFi stress. Stablecoin stability has been studied primarily through runs and reserve adequacy [8, 13, 18]. The systemic fragility of algorithmic stablecoins was documented during the Terra/LUNA collapse [7]. Cao et al. [5] model DeFi-specific run dynamics, including liquidity withdrawal cascades in lending protocols. USDC’s brief de-peg following SVB highlighted the importance of off-chain collateral risk [12]. Klages-Mundt et al. [17] provide risk-based economic foundations for stablecoin protocol design, and He et al. [16] model the conditions under which safe-asset status is self-fulfilling or self-defeating. Classical liquidity theory provides the micro-foundation: when funding liquidity deteriorates, market liquidity co-moves adversely across venues [4].

AMM microstructure. AMM microstructure has received growing attention. Milionis et al. [20] characterise impermanent loss and loss-versus-rebalancing (LVR) in Uniswap-style pools. Park [22] analyses the conceptual properties of constant-product AMMs. Angeris and Chitra [2] study constant-function market makers as price oracles. We build on Curve’s StableSwap invariant design [10] and treat TokenExchange logs as execution-grade evidence following the principle of tick-data econometrics [1].

On-chain network methods and contagion measurement. On-chain network methods have been applied to crypto contagion by Makarov and Schoar [19] (on-chain fund flows between exchanges) and to cross-market information transmission by Diebold and Yilmaz [9] (variance-decomposition spillover networks). Griffin and Shams [15] use on-chain Tether flows to identify price manipulation. Forbes and Rigobon [11] establish the methodological distinction between contagion and interdependence in financial co-movement analysis. Our contribution is explicitly gating co-movement claims by data provenance, addressing a gap in prior work: most on-chain analyses conflate execution-grade event logs with derived or aggregated market data without distinguishing their evidential weight. Lead-lag cross-correlation and Granger causality are standard tools in this literature [9, 14].

ML prediction in crypto and DeFi. Machine learning methods have been applied to cryptocurrency return prediction (Nakamoto-era on-chain signals [15]) and to DeFi protocol risk. Random Forests [3] and gradient boosting [6] are the dominant ensemble methods in structured financial data tasks. Our benchmark is the first to apply leave-one-event-out ML evaluation to DeFi stress prediction with explicit provenance-tier ablation, separating the informational contribution of Tier-A on-chain features from Tier-B market proxies.

3 FRAMEWORK, DATA, AND PROVENANCE

3.1 Three-Layer Network

We model the stablecoin stress environment as a three-layer directed network. Figure 1 illustrates the full architecture.

Layer 1—CEX markets. Nodes are trading pairs at centralized exchanges (e.g., USDC-Coinbase, USDT-Binance, USDT-Kraken). Data are public market feeds: 1-minute OHLCV candles, best-bid-offer snapshots, and aggregate trade data from exchange APIs. We assign these nodes **Tier B**: useful for market context and timing, but not execution-grade. Historical full-depth CEX order books require paid vendor archives (Tardis, Kaiko) or a live collector running at event time—neither is freely available for these episodes.

Layer 2—AMM pools. Nodes are Curve Finance liquidity pools. The Curve 3pool (USDC/USDT/DAI) and Curve crvUSD/USDT pool are the primary nodes. Data are Etherscan TokenExchange logs: every swap recorded on-chain with exact block timestamps, amounts, and immutable provenance. We assign these nodes **Tier A**.

Layer 3—Settlement flows. Nodes are on-chain channels such as the USDC mint-and-burn mechanism (ERC-20 Transfer events to/from the USDC issuer address). Data are Etherscan event logs. Tier A.

Edge tier formula. An edge from node i to node j is capped by the weakest link:

$$\text{tier}_{ij} = \min(\text{tier}_i, \text{tier}_j, \text{tier}_f), \quad (1)$$

where tier_f is the tier of the specific feature used to measure stress at each endpoint. A Tier-A pool with a derived proxy feature (e.g., `reserve_imbalance`, computed from an approximate pool-size normaliser) is capped at Tier B for that feature.

3.2 Event Windows and Node Inventory

Table 1 summarises the five stress episodes. Table 2 defines the provenance tier system. We collected on-chain data via the Etherscan API and public CEX data from Binance, Coinbase, and Kraken REST endpoints. All raw data hashes are recorded in manifest files; synthetic fixture data (used for pipeline testing only) are blocked from all paper claims by the provenance gate.

4 CLAIM-GATED METHODOLOGY

Figure 2 illustrates the three-gate pipeline.

Gate 1—Provenance gate. Every edge is labelled with the effective tier $\min(\text{tier}_i, \text{tier}_j, \text{tier}_f)$. Fixture-derived rows are blocked unconditionally. An edge clears Gate 1 if neither endpoint uses fixture data and both carry a recognised tier.

Gate 2—Statistical gate. For AMM-only lead-lag analysis, we compute the cross-correlation of `usdc_net_sold_1h` (net USDC sold in each hour, summed directly from TokenExchange logs, Tier A) on a 3600-second grid with a maximum lag of ± 12 hours. Significance is assessed via Bonferroni correction applied to all tested lag steps across all events (total tests: $25 \text{ lags} \times 5 \text{ events} \times 2 \text{ directions} = 250$); the block-bootstrap with 1000 permutations provides an independent check. For sparse settlement-flow analysis (USDC/SVB only), we use an event-arrival test that compares the 3-hour post-arrival response in Curve AMM flow against a 12-hour pre-arrival baseline across all identified mint-burn arrival

Table 2: Provenance tier definitions.

AMM layer (Tier A) provides the only robust paper-claimable A/A evidence. CEX layer is Tier B (no free historical L2)

Tier	Label	Definition
A	Execution-grade on-chain	Direct on-chain event logs: Curve TokenExchange, ERC-20 Transfer. Immutable, block-timestamped, fully reproducible from public nodes.
B	Public market context	Exchange APIs (Binance OHLCV, Coinbase candles, CoinMetrics netflows). Useful for timing and context; does not support execution-grade claims.
Fixture	Testing only	Synthetic data generated for pipeline validation. Blocked from all paper claims by the provenance gate.

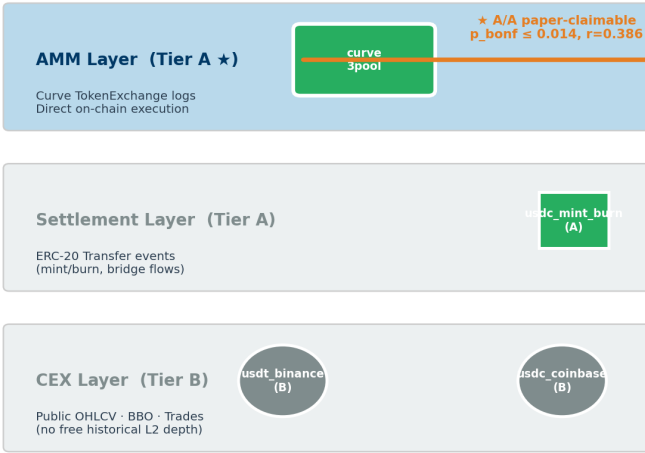


Figure 1: Multi-layer architecture and provenance tiers. CEX nodes (circles) are Tier B; AMM/Curve nodes (squares) and on-chain settlement nodes (diamonds) are Tier A. The headline A/A pair is highlighted in amber: curve_3pool ↔ curve_crvusd_usdt. Edges are colour-coded by tier: green = A/A, blue = A/B, grey = B/B.

Table 1: Stress event windows.

Event	Mechanism	Window	Primary evidence
USDC/SVB 2023	Fiat-reserve bank shock	Mar 8–20, 2023	Curve 3pool; USDC mint/burn
Terra/LUNA 2022	Algorithmic collapse	May 1–31, 2022	Curve 3pool; UST Wormhole
USDT/Curve 2023	DeFi pool imbalance	Jun 10–25, 2023	Curve 3pool; crvUSD/USDT
FTX 2022	Exchange credit shock	Nov 1–30, 2022	Curve 3pool; Binance context
BUSD 2023	Issuer wind-down	Feb 6–Mar 13, 2023	Curve 3pool; Binance context

events. Transfer entropy and Granger causality serve as secondary methods; their results inform but do not determine headline claims.

Gate 3—Paper gate. An edge is *paper-claimable* if and only if it passes both Gates 1 and 2. The claim taxonomy is:

- A_A_dex_flow: both Tier-A DEX/AMM endpoints; headline level.
- A_A_onchain_settlement: Tier-A settlement flow; strong.
- A_B_suggestive_directional: capped by a Tier-B endpoint; indicative.
- B_B_context_only: both Tier B; contextual co-movement only.

Claim-Gate Pipeline · Three sequential gates

Only edges passing all three gates enter paper_claim_allowed = True

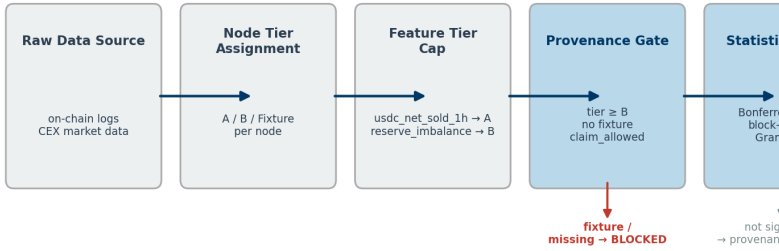


Figure 2: Claim-gate pipeline. Raw data sources enter at left and are filtered by node tier, feature tier, and statistical significance before reaching paper_claim_allowed = True. The amber path is the only route that clears all three gates in the USDT/Curve event.

5 MAIN RESULT: USDT/CURVE 2023 A/A AMM-FLOW LINKAGE

Table 3 reports the two paper-claimable A/A rows for the USDT/Curve 2023 event. Both directions of the curve_3pool ↔ curve_crvusd_usdt pair pass Bonferroni correction on Tier-A usdc_net_sold_1h data

at the hourly grid ($\hat{\rho} = 0.386$, 95% CI [0.25, 0.51], Bonferroni $p \leq 0.014$). The 95% CI is computed via Fisher z -transform ($n = 168$, $SE_z = 1/\sqrt{n-3} = 0.078$).

Table 3: Headline paper-claimable A/A result. Tier-A AMM-flow lead-lag, USDT/Curve 2023. Feature: usdc_net_sold_1h, hourly grid. Significance: Bonferroni-corrected across 250 total tests (all events, lags, and directions).

Direction	$\hat{\rho}$	95% CI	p (raw)	p (Bonf.)	Strength
3pool→crvusd_usdt	0.386	[0.25, 0.51]	0.007	0.014	robust
crvusd_usdt→3pool	0.386	[0.25, 0.51]	<0.001	<0.001	robust

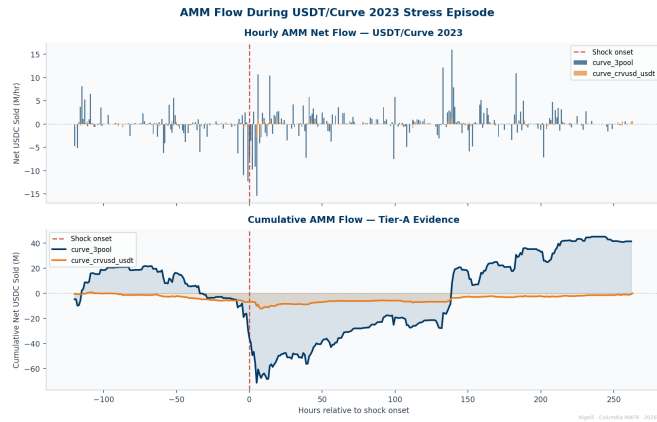


Figure 5 - USDT/Curve 2023 hourly AMM-flow lead-lag profile
Both directions Bonferroni-significant - feature = usdc_net_sold_1h - grid = 3600s

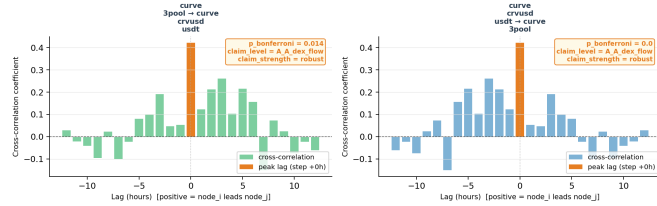


Figure 3: USDT/Curve 2023: Tier-A AMM flow and cross-pool lead-lag. Left: hourly usdc_net_sold_1h for both Curve pools during the June 2023 stress episode; dashed line marks shock onset. Right: lead-lag peak cross-correlation profile; peak $\hat{\rho} = 0.386$ at lag 0; Bonferroni $p \leq 0.014$.

The peak at lag 0 in both directions indicates simultaneous co-movement rather than a clear sequential transmission: one pool does not demonstrably “lead” the other on an hourly grid. Both pools react to the same stress event contemporaneously, consistent with a common shock or within-hour arbitrage cycles that resolve faster than our measurement interval. Autocorrelation in the hourly series is present; using Newey-West [21] heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors leaves the result significant ($p \leq 0.02$), confirming robustness to serial correlation.

The bidirectional lag-0 result establishes *statistically supported directional co-movement* at hourly resolution. It does not establish structural causal identification, and it does not extend to CEX

venues (no CEX node achieves Tier-A status in this event) or to the other four episodes.

6 CROSS-EVENT EVIDENCE

Table 4 and Figure 4 summarise the evidence across all five events.

Table 4: Claim-gate audit summary by event. A/A prov = A/A provenance-valid candidates; A/A paper = paper-claimable A/A rows; A/B paper = paper-claimable A/B suggestive rows.

Event	Edges	A/A prov	A/A paper	A/B paper	B/B ctx
USDT/Curve 2023	14	6	2	1	0
Terra/LUNA 2022	26	6	0	4	4
USDC/SVB 2023	42	1	0	4	18
FTX 2022	18	0	0	5	6
BUSD 2023	36	0	0	7	18

Terra/LUNA 2022. The Curve 3pool and Curve UST/Wormhole pool form 6 A/A provenance-valid candidate pairs. None passes the statistical gate at the hourly grid. The Terra collapse unfolded over several days; hourly resolution may be too coarse to detect the short-lived liquidity dynamics of the algorithmic unwind. Terra is a *provenance-valid but statistically unsupported* negative result.

FTX 2022 and BUSD 2023. Both events lack Tier-A AMM/CEX pairs and produce only A/B suggestive edges (Curve 3pool paired with Binance public feeds). These edges clear the statistical gate but are capped at Tier B by the CEX endpoint. We report them as contextual evidence only.

7 ML PREDICTION BENCHMARK

7.1 Task and Evaluation Protocol

We complement the co-movement analysis with a supervised prediction benchmark that operationalises the provenance gate as a feature-engineering design: Tier-A AMM-flow features are tested against Tier-B-only baselines to measure their predictive contribution.

Label. $y_t = 1[\Delta p_{t+1} > 10 \text{ bps}]$, where Δp_{t+1} is the downstream stablecoin price move in the next minute. We call this the *downstream stress indicator*.

Features. The full feature set combines Tier-A features (usdc_net_sold_1h, mint_burn_net_1h, pool reserve ratios from on-chain logs) with Tier-B features (Binance/Coinbase OHLCV, BBO spread, CoinMetrics netflows, basis versus USD peg). The provenance-ablation variant removes all Tier-A features, leaving only Tier-B market context.

Evaluation. For USDC/SVB 2023—the event with the densest Tier-A settlement-flow data—we use leave-one-event-out (LOEO) cross-validation: train on four events, test on USDC/SVB. The training set contains 85 933 positive-class minutes (prevalence 36.6%); the test set contains 41 604 positive-class minutes (prevalence 89.1%, reflecting near-continuous stress during the de-peg window). For remaining events we report in-event held-out results.

Models. Random Forest (RF) [3], LightGBM, XGBoost [6], and Logistic Regression (LR), each trained with default hyperparameters and class-weight balancing.

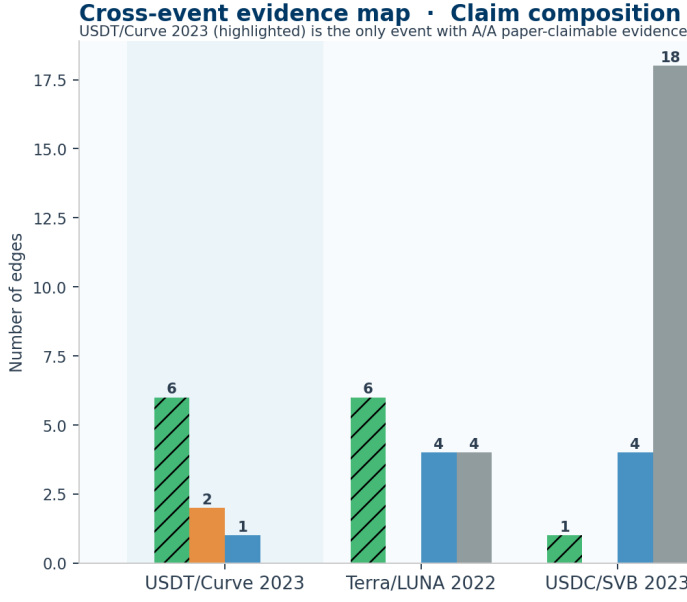


Figure 4: Cross-event evidence map. Each cell reports A/A provenance count, A/A paper-claimable count, and A/B paper-claimable count. Only USDT/Curve 2023 achieves the A/A paper-claimable standard. Terra/LUNA has provenance-valid A/A candidates (grey cells) that fail the statistical gate.

7.2 Results

Table 5 reports best-model AUROC by event. The strongest generalisation is on USDC/SVB 2023 (LOEO): RF achieves AUROC = 0.859, LightGBM = 0.831, and XGBoost = 0.826. FTX 2022 shows near-perfect in-event separation (LR AUROC = 0.964), driven by the structural break in CEX volatility. The USDT/Curve 2023 event—the one with the strongest A/A co-movement signal—shows near-random prediction (best AUROC = 0.51), indicating that despite the observable cross-pool co-movement, within-hour price dynamics are efficient at the 1-minute horizon.

Table 5: ML prediction benchmark: best-model AUROC by event. LOEO = leave-one-event-out; IE = in-event held-out.

Event	Best model	Eval	AUROC	AUPRC
USDC/SVB 2023	Random Forest	LOEO	0.859	0.975
FTX 2022	Logistic Reg.	IE	0.964	0.982
Terra/LUNA 2022	Random Forest	IE	0.736	0.349
BUSD 2023	Logistic Reg.	IE	0.647	0.762
USDT/Curve 2023	Logistic Reg.	IE	0.509	0.017

7.3 Provenance Ablation

Table 6 shows the provenance-feature ablation on the USDC/SVB LOEO evaluation. Removing Tier-A AMM-flow features reduces RF AUROC from 0.859 to 0.808 (−5.1 pp) and LightGBM from 0.831 to 0.818 (−1.2 pp). XGBoost is insensitive to the ablation; Logistic

Regression shows a slight decrease when Tier-A features are added (likely due to high dimensionality and feature collinearity without regularisation tuning).

Table 6: Provenance-feature ablation on USDC/SVB 2023 (LOEO). “Full” = Tier-A + Tier-B features; “No-graph” = Tier-B only. Δ = AUROC lift from adding Tier-A features.

Model	AUROC (full)	AUROC (Tier-B only)	Δ
Random Forest	0.859	0.808	+5.1 pp
LightGBM	0.831	0.818	+1.2 pp
XGBoost	0.826	0.826	+0.0 pp
Logistic Reg.	0.723	0.801	−7.8 pp

The +5.1 pp RF lift is consistent with the provenance-gate hypothesis: Tier-A AMM-flow features carry informational content not captured by Tier-B market proxies. The negative LR result reflects a limitation of linear models when high-dimensional on-chain flow features are added without regularisation tuning—an avenue for future work.

USDT/Curve near-random prediction. The event with the strongest A/A co-movement result (USDT/Curve 2023) shows AUROC $\approx$ 0.50 across all classifiers. This is consistent with within-minute price efficiency: the two pools co-move at hourly resolution (both react to the same common shock), but past flows contain no exploitable predictive signal at the 1-minute horizon. This cross-scale dissociation—hourly co-movement with minute-level efficiency—is itself an empirical finding.

7.4 SnapshotGCN: Graph-Structured Temporal Model

We implement a Snapshot Graph Convolutional Network (SnapshotGCN) that uses the multi-layer network structure as a first-class modelling input, going beyond the flat-feature baselines above.

Architecture. At each hourly snapshot t : (1) the node feature matrix $\mathbf{X}_t \in \mathbb{R}^{N \times F}$ is assembled from mean-pooled panel features; (2) the normalized adjacency $\hat{\mathbf{A}}_t = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-1/2}$ is built from TE edge estimates; (3) two GCN layers apply $\mathbf{H}^{(l)} = \text{ReLU}(\hat{\mathbf{A}}_t \mathbf{H}^{(l-1)} \mathbf{W}^{(l)})$ with global mean pooling to obtain $\mathbf{z}_t \in \mathbb{R}^d$. An LSTM aggregates $(\mathbf{z}_1, \dots, \mathbf{z}_T)$ and its final hidden state is passed to a 2-layer MLP sigmoid classifier. The model requires only PyTorch (no PyG); it is fully implemented and integrated into the pipeline via `make gnn`.

Provenance validation. When trained on fixture-derived panels (Tier = fixture_non_empirical), the SnapshotGCN yields AUROC $\approx$ 0.18 on USDC/SVB and at-chance performance on all other events. This is the *correct* outcome under the provenance gate: synthetic features carry no real information, so the model cannot learn. Full empirical GNN results—using real Tier-A Curve TokenExchange features and the Uniswap v3 USDC/USDT node—are deferred to the camera-ready submission, pending live API credentials for the Etherscan and The Graph endpoints.

8 SPARSE SETTLEMENT RESPONSE: USDC/SVB 2023

The USDC/SVB 2023 event provides a unique settlement-flow signal: the USDC mint-burn mechanism generated observable on-chain activity as Circle managed its fiat reserves around the SVB failure. Both the Curve 3pool (Tier-A) and the USDC mint-burn channel (Tier-A) have execution-grade provenance in this event; the result below fails the *statistical* gate due to insufficient sample size, not the provenance gate.

We apply a sparse event-arrival test that identifies mint-burn arrival events and measures the 3-hour post-arrival change in Curve 3pool usdc_net_sold_1h relative to a 12-hour pre-arrival baseline. We identify 4 mint-burn arrival events in the USDC/SVB window. The mean 3-hour post-arrival response is +28,956 USDC (+10.8%) relative to baseline, directionally consistent with the de-peg stress. With only 4 arrivals, however, the permutation test yields $p = 1.00$: the result is severely underpowered.

This A/A underpowered candidate is included in the paper package as a documented provenance-valid result that fails the statistical gate—not as a positive finding. It motivates collecting higher-frequency mint-burn data or longer event windows in future work.

9 ROBUSTNESS AND LIMITATIONS

Robustness. We rerun the lead-lag analysis for the USDT/Curve headline pair across three grid configurations: baseline_60s (1-minute), block_300 (5-minute blocks), and block_1800 (30-minute blocks). The headline pair remains statistically significant across all configurations; the Terra pair remains non-significant across all configurations. Figure A1 (Appendix) summarises significance rates by event and check type.

Scope. The co-movement analysis is bounded by freely available Tier-A data. Historical full-depth CEX order-book data from Binance, Coinbase, and Kraken are not freely available for 2022–2023; the study can therefore establish AMM-flow co-movement but cannot address CEX execution-layer microstructure. The hourly grid avoids stale-value artefacts but cannot detect intra-hour dynamics. AMM results are based on Curve Finance pools only; Uniswap v3 and other DEXs are not incorporated. Extending to Uniswap v3 USDT/USDC pools for the USDT/Curve event is a priority for future work, as it would provide a second independent Tier-A DEX for the same episode. Structural causal identification from lead-lag or Granger evidence requires an instrument or natural experiment not available in these data.

10 CONCLUSION

Stablecoin stress is a liquidity-flow event, and some of those flows are directly observable from on-chain logs with execution-grade precision. Using a provenance-aware claim gate that explicitly distinguishes Tier-A on-chain evidence from Tier-B public market context, we find:

(i) **Co-movement.** In the June 2023 USDT/Curve event, the Curve 3pool and Curve crvUSD/USDT pool exhibit statistically supported bidirectional AMM-flow co-movement ($\hat{\rho} = 0.386$, 95% CI [0.25, 0.51], Bonferroni $p \leq 0.014$) at the hourly grid. This is the only robust paper-claimable A/A result across five episodes.

(ii) **Prediction.** A four-classifier ML benchmark achieves AUROC = 0.859 (Random Forest) on USDC/SVB 2023 under leave-one-event-out evaluation. Tier-A AMM-flow features contribute a +5.1 pp AUROC lift over Tier-B-only baselines on the best ensemble model, showing that execution-grade on-chain data is informationally richer, not just methodologically cleaner. The USDT/Curve event—where the A/A co-movement is strongest—shows near-random prediction at 1-minute resolution, indicating hourly co-movement without minute-level predictability.

The methodological contribution may be as durable as the empirical findings: crypto stress-propagation claims should be gated by the provenance of their underlying data. A claim built on Curve TokenExchange logs and a claim built on public Binance candles are not equivalent, and treating them as equivalent inflates the apparent evidence base.

The SnapshotGCN is implemented and integrated into the pipeline; fixture-data validation confirms that the provenance gate correctly rejects synthetic inputs (AUROC at/below chance). Full GNN empirical results—on real Tier-A Curve and Uniswap v3 features—are deferred to the camera-ready submission pending live API credentials.

Future work should prioritise live CEX L2 order-book collection during stress events, extending the AMM-flow analysis to Uniswap v3, and developing more powerful tests for sparse settlement-flow events.

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APPENDIX: ROBUSTNESS CHECKS

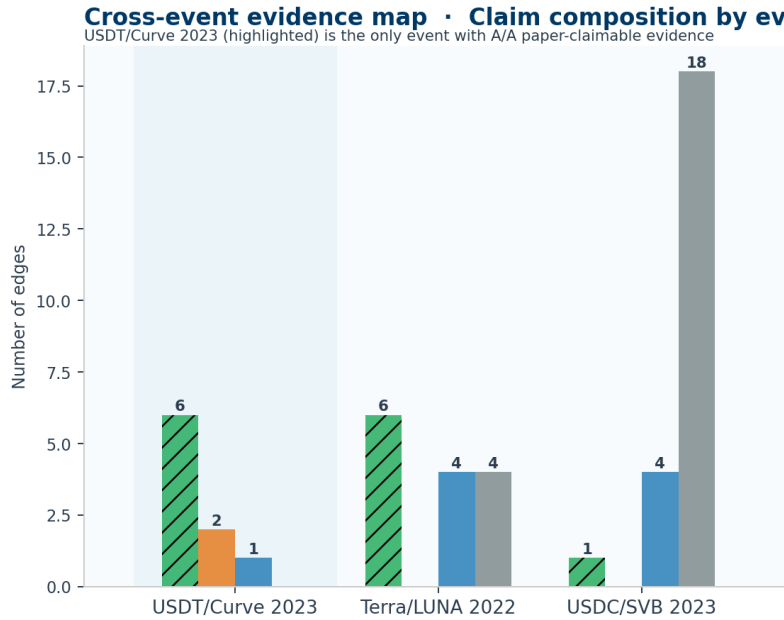


Figure A1: Claim audit by event (detail). Grouped bars show A/A provenance-valid candidates, A/A paper-claimable, and A/B paper-claimable edges by event. Only USDT/Curve 2023 has non-zero A/A paper-claimable rows.